Q1: 16 March (Shift 1) - Single Correct

If $\mathrm{y}=\mathrm{y}(\mathrm{x})$ is the solution of the differential equation, $\frac{\mathrm{dy}}{\mathrm{dx}}+2\mathrm{y}\tan\mathrm{x}=\sin\mathrm{x}, \mathrm{y}\left(\frac{\pi}{3}\right)=0$, then the maximum value of the function $\mathrm{y}(\mathrm{x})$ over $\mathbb{R}$ is equal to :

(1) 8

(2) $\frac{1}{2}$

(3) $-\frac{15}{4}$

(4) $\frac{1}{8}$

Q2: 16 March (Shift 2) - Single Correct

If $\mathrm{y}=\mathrm{y}(\mathrm{x})$ is the solution of the differential equation $\frac{\mathrm{dy}}{\mathrm{dx}}+(\tan\mathrm{x})\mathrm{y}=\sin\mathrm{x}, 0\leq\mathrm{x}\leq\frac{\pi}{3}$, with $\mathrm{y}(0)=0$, then $\mathrm{y}\left(\frac{\pi}{4}\right)$ equal to :

(1) $\frac{1}{4}\log_{\mathrm{e}}2$

(2) $\left(\frac{1}{2\sqrt{2}}\right)\log_{\mathrm{e}}2$

(3) $\log_{\mathrm{e}}2$

(4) $\frac{1}{2}\log_{\mathrm{e}}2$

Q3: 17 March (Shift 1) - Single Correct

Which of the following is true for $y(x)$ that satisfies the differential equation $\frac{\mathrm{dy}}{\mathrm{dx}}=\mathrm{xy}-1+\mathrm{x}-\mathrm{y};\mathrm{y}(0)=0$

(1) $y(1)=e^{-\frac{1}{2}}-1$

(2) $y(1)=e^{\frac{1}{2}}-e^{-\frac{1}{2}}$

(3) $y(1)=1$

(4) $y(1)=e^{\frac{1}{2}}-1$

Q4: 17 March (Shift 2) - Single Correct

Let $y=y(x)$ be the solution of the differential equation

$$\cos x(3\sin x+\cos x+3)dy = (1+y\sin x(3\sin x+\cos x+3))dx$$

$0 \leq \mathrm{x} \leq \frac{\pi}{2}, \mathrm{y}(0)=0$. Then , $\mathrm{y}\left(\frac{\pi}{3}\right)$ is equal to:

(1) $2\log_{\mathrm{e}}\left(\frac{2\sqrt{3}+9}{6}\right)$

(2) $2\log_{e}\left(\frac{2\sqrt{3}+10}{11}\right)$

(3) $2\log_{\mathrm{e}}\left(\frac{\sqrt{3}+7}{2}\right)$

(4) $2\log_{e}\left(\frac{3\sqrt{3}-8}{4}\right)$

Q5: 17 March (Shift 2) - Single Correct

If the curve $y=y(x)$ is the solution of the differential equation

$2\left(x^2+x^{5/4}\right)dy-y\left(x+x^{1/4}\right)dx=2x^{9/4}dx, x>0$

which passes through the point $\left(1, 1-\frac{4}{3}\log_{\mathrm{e}}2\right)$, then the value of $\mathrm{y}(16)$ is equal to :

(1) $4\left(\frac{31}{3}+\frac{8}{3}\log_{\mathrm{e}}3\right)$

(2) $\left(\frac{31}{3}+\frac{8}{3}\log_{\mathrm{e}}3\right)$

(3) $4\left(\frac{31}{3}-\frac{8}{3}\log_{\mathrm{e}}3\right)$

(4) $\left(\frac{31}{3}-\frac{8}{3}\log_{\mathrm{e}}3\right)$

Q6: 18 March (Shift 1) - Single Correct

The differential equation satisfied by the system of parabolas $\mathrm{y}^2=4\mathrm{a}(\mathrm{x}+\mathrm{a})$ is:

(1) $y\left(\frac{dy}{dx}\right)^2-2x\left(\frac{dy}{dx}\right)-y=0$

(2) $y\left(\frac{dy}{dx}\right)^2-2x\left(\frac{dy}{dx}\right)+y=0$

(3) $y\left(\frac{dy}{dx}\right)^2+2x\left(\frac{dy}{dx}\right)-y=0$

(4) $\mathrm{y}\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)+2\mathrm{x}\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)-\mathrm{y}=0$

Q7: 18 March (Shift 2) - Single Correct

Let $y=y(x)$ be the solution of the differential equation $\frac{\mathrm{dy}}{\mathrm{dx}}=(\mathrm{y}+1)\left((\mathrm{y}+1) \mathrm{e}^{\mathrm{x}^2 / 2}-\mathrm{x}\right), 0<\mathrm{x}<2.1$, with $\mathrm{y}(2)=0$. Then the value of $\frac{\mathrm{dy}}{\mathrm{dx}}$ at $\mathrm{x}=1$ is equal to:

(1) $\frac{-\mathrm{e}^{3 / 2}}{\left(\mathrm{e}^2+1\right)^2}$

(2) $-\frac{2 \mathrm{e}^2}{\left(1+\mathrm{e}^2\right)^2}$

(3) $\frac{\mathrm{e}^{5 / 2}}{\left(1+\mathrm{e}^2\right)^2}$

(4) $\frac{5 \mathrm{e}^{1 / 2}}{\left(\mathrm{e}^2+1\right)^2}$

Answer Key

Q1 (4)	**Q2** (2)	**Q3** (1)	**Q4** (2)
Q5 (3)	**Q6** (3)	**Q7** (1)	

Q1 (4)

$\frac{dy}{dx}+2y\tan x=\sin x$

$\mathrm{I.F.}=e^{\int 2\tan xdx}=e^{2\mathrm{tnsecx}}$

$\mathrm{I.F.}=\sec^2 x$

$y.\left(\sec^2 x\right)=\int \sin x\cdot\sec^2 xdx$

$y.\left(\sec^2 x\right)=\int \sec x\tan xdx$

$y.\left(\sec^2 x\right)=\sec x+C$

$x=\frac{\pi}{3};y=0$

$\Rightarrow\quad C=-2$

$\Rightarrow\quad y=\frac{\sec x-2}{\sec^2 x}=\cos x-2\cos^2 x$

$y=t-2t^2\Rightarrow\frac{dy}{dt}=1-4t=0\Rightarrow t=\frac{1}{4}$

$\therefore\quad \max=\frac{1}{4}-\frac{1}{8}=\frac{2-1}{8}=\frac{1}{8}$

Q2 (2)

$\frac{dy}{dx}+(\tan x)y=\sin x;0\leq x\leq\frac{\pi}{3}$

$\mathrm{I.F.}=e^{\int\tan xdx}=e^{\ell n\sec x}=\sec x$

$y\sec x=\int\tan xdx$

$y\sec x=\int\tan xdx$

$y\sec x=\ell n|\sec x|+C$

$x=0,y=0\quad\Rightarrow\quad\therefore c=0$

$y\sec x=\ell n|\sec x|$

$y=\cos x\cdot\ln|\sec x|$

$y|_{x=\frac{\pi}{4}}=\left(\frac{1}{\sqrt{2}}\right)\cdot\ell n\sqrt{2}$

$y|_{x=\frac{\pi}{4}}=\frac{1}{2\sqrt{2}}\log_e 2$

Q3 (1)

$\frac{dy}{dx}=(1+y)(x-1)$

$\frac{dy}{(y+1)}=(x-1)dx$

Integrate $\ln(y+1)=\frac{x^2}{2}-x+c$

$(0,0)\Rightarrow c=0\Rightarrow y=e^{\left(\frac{x^2}{2}-x\right)}-1$

Q4 (2)

Sol. $\cos x(3\sin x+\cos x+3)\ dy$

$$=(1+y\sin x(3\sin x+\cos x+3))dx$$
$$\frac{dy}{dx}-(\tan x)y=\frac{1}{(3\sin x+\cos x+3)\cos x}$$
$$\text{I.F.}=e^{\int-\tan x dx}=e^{(n\cos x|}=|\cos x|$$
$$=\cos x\forall x\in\left[0,\frac{\pi}{2}\right)$$

Solution of D. E

$$y(\cos x)=\int(\cos x)\cdot\frac{1}{\cos x(3\sin x+\cos x+3)}dx+C$$

$$y(\cos x)=\int\frac{dx}{3\sin x+\cos x+3}dx+C$$
$$y(\cos x)=\int\frac{\left(\sec^2\frac{x}{2}\right)}{2\tan^2\frac{x}{2}+6\tan\frac{x}{2}+4}dx+C$$

Now $I_1=\int\frac{\left(\sec^2\frac{x}{2}\right)}{2\left(\tan^2\frac{x}{2}+3\tan\frac{x}{2}+2\right)}dx+C$

Put $\tan\frac{x}{2}=t\Rightarrow\frac{1}{2}\sec^2\frac{x}{2}dx=dt$

$I_1=\int\frac{dt}{t^3+3t+2}=\int\frac{dt}{(t+2)(t+1)}$

$=\int\left(\frac{1}{t+1}-\frac{1}{t+2}\right)dt$

$=\ell n\left|\left(\frac{t+1}{t+2}\right)\right|=\ell n\left|\left(\frac{\tan\frac{x}{2}+1}{\tan\frac{x}{2}+2}\right)\right|$

So solution of D. E

$y(\cos x)=\ell n\left|\frac{1+\tan\frac{x}{2}}{2+\tan\frac{x}{2}}\right|+C$

$\Rightarrow y(\cos x)=\ell n\left(\frac{1+\tan\frac{x}{2}}{2+\tan\frac{x}{2}}\right)+C\quad$ for $0\leq x<\frac{\pi}{2}$

Now, it is given $y(0)=0$

$\Rightarrow 0=\ell n\left(\frac{1}{2}\right)+C\quad\Rightarrow\quad C=\ell n2$

$\Rightarrow y(\cos x)=\ell n\left(\frac{1+\tan \frac{x}{2}}{2+\tan \frac{x}{2}}\right)+\ell n 2$

For $x=\frac{\pi}{3}$

$\mathrm{y}\left(\frac{1}{2}\right)=\ln \left(\frac{1+\frac{1}{\sqrt{3}}}{2+\frac{1}{\sqrt{3}}}\right)+\ell n 2$

$\mathrm{y}=2 \ell \mathrm{n}\left(\frac{2 \sqrt{3}+10}{11}\right)$

Q5 (3)

$\frac{\mathrm{dy}}{\mathrm{dx}}-\frac{\mathrm{y}}{2 \mathrm{x}}=\frac{\mathrm{x}^{9 / 4}}{\mathrm{x}^{5 / 4}\left(\mathrm{x}^{3 / 4}+1\right)}$

$\mathrm{IF}=\mathrm{e}^{-\int \frac{\mathrm{dx}}{2 \mathrm{~d}}}=\mathrm{e}^{-\frac{1}{2} \ln \mathrm{x}}=\frac{1}{\mathrm{x}^{1 / 2}}$

$\mathrm{y} \cdot \mathrm{x}^{-1 / 2}=\int \frac{\mathrm{x}^{9 / 4} \cdot \mathrm{x}^{-1 / 2}}{\mathrm{x}^{5 / 4}\left(\mathrm{x}^{3 / 4}+1\right)} \mathrm{dx}$

$\int \frac{\mathrm{x}^{1 / 2}}{\left(\mathrm{x}^{3 / 4}+1\right)} \mathrm{dx}$

$x=t^4 \Rightarrow d x=4 t^3 d t$

$\int \frac{t^2 \cdot 4 t^3 d t}{\left(t^3+1\right)}$

$4 \int \frac{t^2\left(t^3+1-1\right)}{\left(t^3+1\right)} d t$

$4 \int t^2 d t-4 \int \frac{t^2}{t^3+1} d t$

$\frac{4 t^3}{3}-\frac{4}{3} \ln \left(t^3+1\right)+C$

$y x^{-1 / 2}=\frac{4 x^{3 / 4}}{3}-\frac{4}{3} \ln \left(x^{3 / 4}+1\right)+C$

$1-\frac{4}{3} \log _e 2=\frac{4}{3}-\frac{4}{3} \log _e 2+C$

$\Rightarrow C=-\frac{1}{3}$

$y=\frac{4}{3} x^{5 / 4}-\frac{4}{3} \sqrt{x} \ln \left(x^{3 / 4}+1\right)-\frac{\sqrt{x}}{3}$

$y(16)=\frac{4}{3} \times 32-\frac{4}{3} \times 4 \ln 9-\frac{4}{3}$

$=\frac{124}{3}-\frac{32}{3} \ln 3=4\left(\frac{31}{3}-\frac{8}{3} \ln 3\right)$

Q6 (3)

$y^2=4 a x+4 a^2$

differentiate with respect to x

$\Rightarrow 2 \mathrm{y} \frac{\mathrm{dy}}{\mathrm{dx}}=4 \mathrm{a}$

$\Rightarrow \mathrm{a}=\left(\frac{\mathrm{y}}{2} \frac{\mathrm{dy}}{\mathrm{dx}}\right)$

so, required differential equation is

$y^2=\left(4\times\frac{y}{2}\frac{dy}{dx}\right)x+4\left(\frac{y}{2}\frac{dy}{dx}\right)^2$

$\Rightarrow \mathrm{y}^2\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)^2+2\mathrm{xy}\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)-\mathrm{y}^2=0$

$\Rightarrow \mathrm{y}\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)^2+2\mathrm{x}\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)-\mathrm{y}=0$

Q7 (1)

Let $\mathrm{y}+1=\mathrm{Y}$

$\therefore \frac{\mathrm{dY}}{\mathrm{dx}}=\mathrm{Y}^2\mathrm{e}^{\frac{x^2}{2}}-\mathrm{xY}$

Put $-\frac{1}{Y}=k$

$\Rightarrow \frac{\mathrm{dk}}{\mathrm{dx}}+\mathrm{k}(-\mathrm{x})=\mathrm{e}^{\frac{\mathrm{x}^2}{2}}$

I.F. $=\mathrm{e}^{-\frac{x^2}{2}}$

$\therefore \mathrm{k}=(\mathrm{x}+\mathrm{c})\mathrm{e}^{\mathrm{x}^2/2}$

Put $k=-\frac{1}{y+1}$

$\therefore \mathrm{y}+1=-\frac{1}{(\mathrm{x}+\mathrm{c})\mathrm{e}^{\mathrm{x}^2/2}}$

when $x=2, y=0$, then $c=-2-\frac{1}{e^2}$

diffentiate equation

(i) & put $x=1$

we get $\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)_{\mathrm{x}=1}=-\frac{\mathrm{e}^{3/2}}{(1+\mathrm{e}^2)^2}$